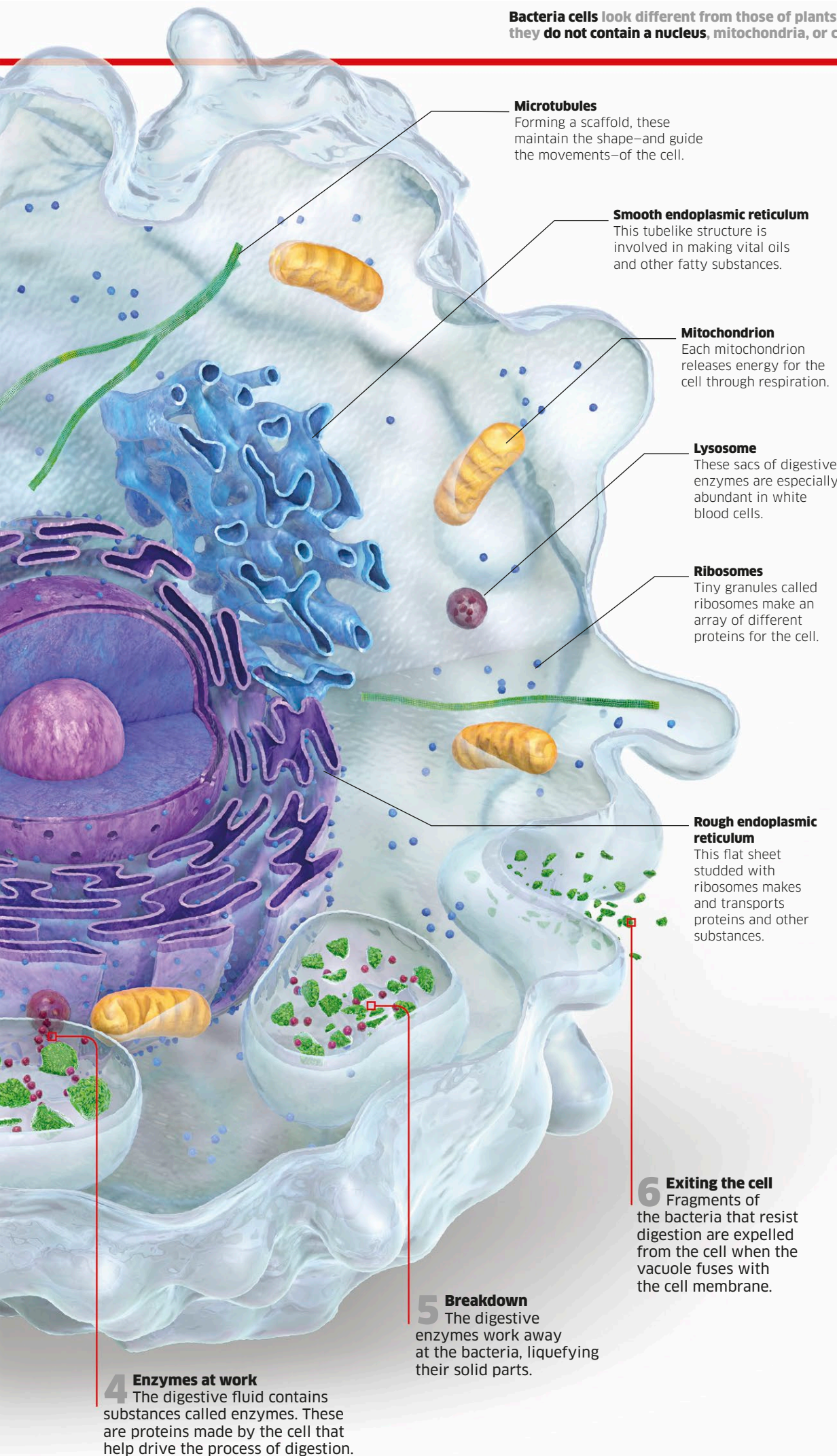




**Microtubules**

Forming a scaffold, these maintain the shape—and guide the movements—of the cell.

**Smooth endoplasmic reticulum**

This tubelike structure is involved in making vital oils and other fatty substances.

**Mitochondrion**

Each mitochondrion releases energy for the cell through respiration.

**Lysosome**

These sacs of digestive enzymes are especially abundant in white blood cells.

**Ribosomes**

Tiny granules called ribosomes make an array of different proteins for the cell.

**Rough endoplasmic reticulum**

This flat sheet studded with ribosomes makes and transports proteins and other substances.

**6 Exiting the cell**

Fragments of the bacteria that resist digestion are expelled from the cell when the vacuole fuses with the cell membrane.

**5 Breakdown**

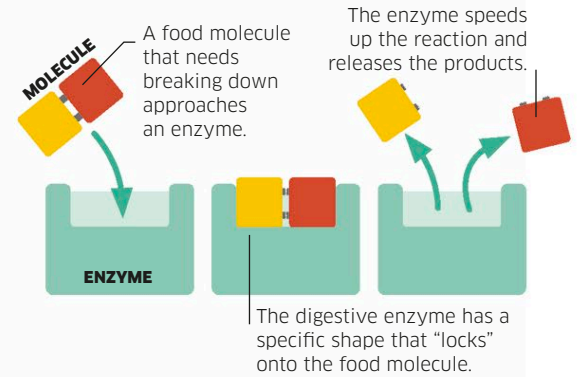
The digestive enzymes work away at the bacteria, liquefying their solid parts.

**4 Enzymes at work**

The digestive fluid contains substances called enzymes. These are proteins made by the cell that help drive the process of digestion.

**Enzymes**

Cells make complex molecules called proteins, many of which work as enzymes. Enzymes are catalysts—substances that increase the rate of chemical reactions and can be used again and again. Each type of reaction needs a specific kind of enzyme.



**Cell variety**

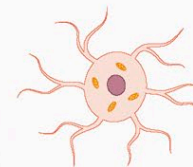
Unlike animal cells, plant cells are ringed by a tough cell wall and many have food-making chloroplasts. Both animals and plants have many specialised cells for different tasks.

**ANIMAL CELLS**



**Fat cell**

Its large droplet of stored fat provides energy when needed.



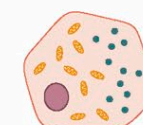
**Bone-making cell**

Long strands of cytoplasm help this cell connect to others.



**Ciliated cell**

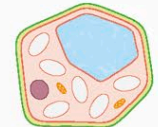
Hairlike cilia waft particles away from airways.



**Secretory cell**

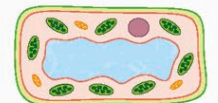
These cells release useful substances, such as hormones.

**PLANT CELLS**



**Starch-storing cell**

Some root cells store many granules of energy-rich starch.



**Leaf cell**

Inside this cell, green chloroplasts make food for the plant.



**Supporting cell**

Thick-walled cells in the stem help support plants.



**Fruit cell**

Its large sap-filled vacuole helps to make a fruit juicy.